

Praktikum Hochfrequenz-Schaltungstechnik WS25/26

Bericht

Name:	Sebastian Grigorevski	Matrikelnr.:	35690104
Versuchsnr.:	3	Datum:	23.12.2025

3.5.1

Figure 1 shows the measured resistance R_s and impedance $|Z|$ of a carbon resistor and a coaxial termination over a frequency range between 1 MHz and 30 MHz. The coaxial termination exhibits behavior very close to the ideal expectation. Both the R_s and the $|Z|$ remain nearly constant at $\sim 50\Omega$ over the entire frequency range. Thus it is well impedance matched.

In contrast, the carbon resistor exhibits a clear frequency dependency. The impedance magnitude increases with frequency and exceeds 51Ω at higher frequencies. This behavior is caused by parasitic inductive effects inherent to the resistor's internal structure and material.

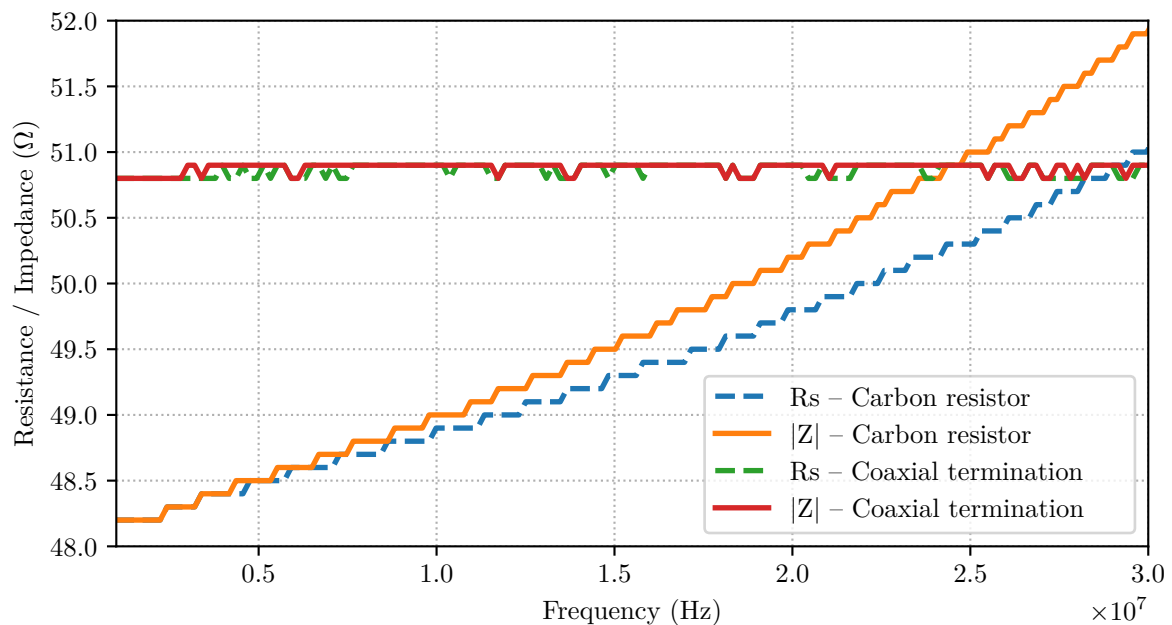


Figure 1: Comparison of resistance and impedance of carbon resistor and coaxial termination

3.5.2

Figure 2 shows the measured impedance magnitude $|Z|$ and phase of the inductor as a function of frequency. At $f = 3.26$ MHz, the inductance is measured at $\sim 50\Omega$. The theoretical value is obtained from $|Z| \approx X_L = 2\pi fL \approx 45\Omega$ with $L=2.2 \mu\text{H}$. Small deviations are caused by series resistance and measurement uncertainty. The impedance reaches $|Z| = 50\Omega$ at low frequency where the phase is still strongly positive ($+88.86^\circ$), confirming inductive behavior. A pronounced maximum of $|Z|$ occurs at ~ 30 MHz, indicating the self-resonant frequency of the inductor due to parasitic capacitance. Therefore, the maximum operating frequency of the inductor is below resonance.

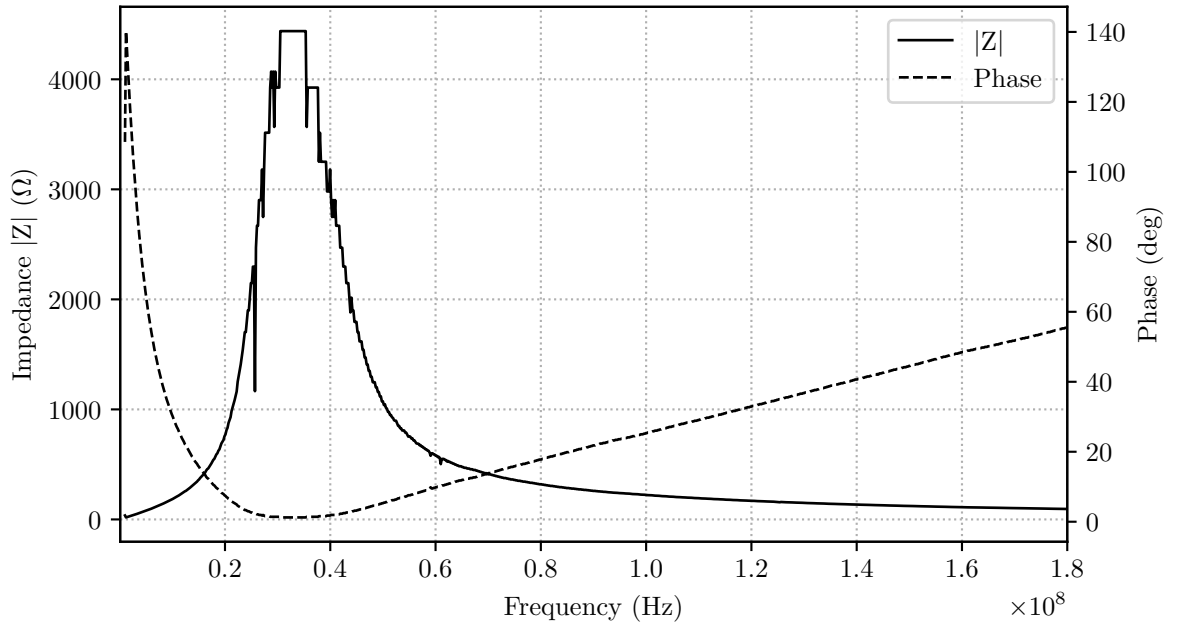


Figure 2: Measured impedance magnitude and phase of an inductor as a function of frequency.

3.5.3

The 3 dB bandwidth is determined from the frequencies where the return loss is 3 dB above the minimum. From the figure 3, a bandwidth of approximately $\Delta f_{3\text{dB}} \approx 540$ Hz is obtained. The quality factor of the quartz resonator is therefore

$$Q = \frac{f_0}{\Delta f_{3\text{dB}}} \approx \frac{9.8283 \text{ MHz}}{540 \text{ Hz}} \approx 18200.45$$

indicating a very high Q and highly frequency-selective resonator [3].

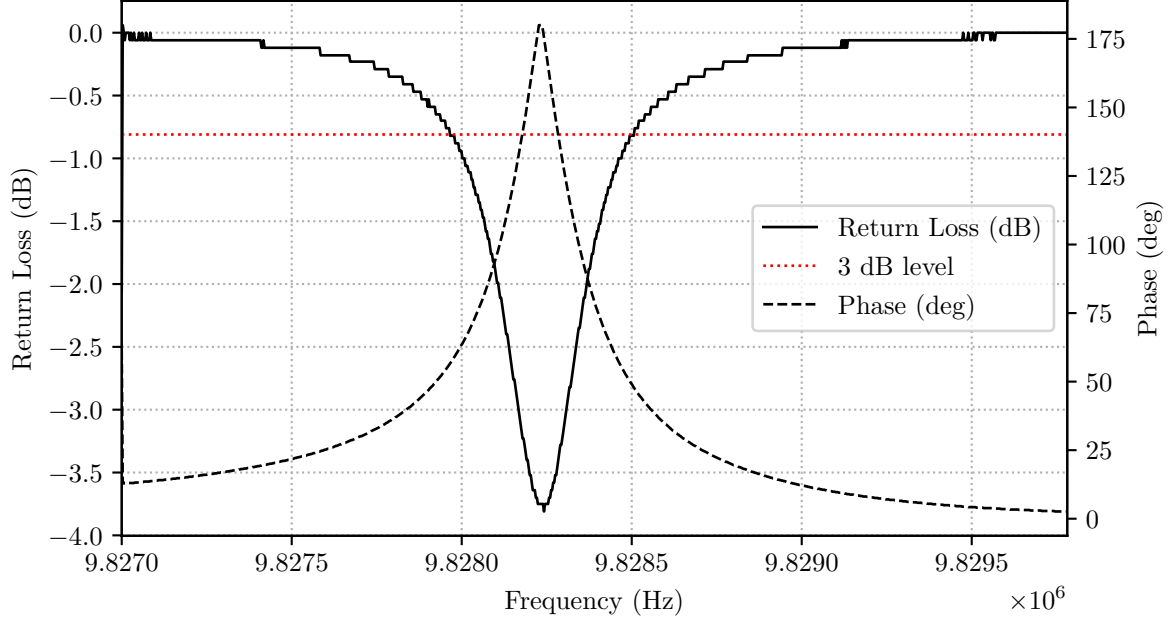


Figure 3: Measured return loss and phase of a quartz resonator as a function of frequency.

3.5.4

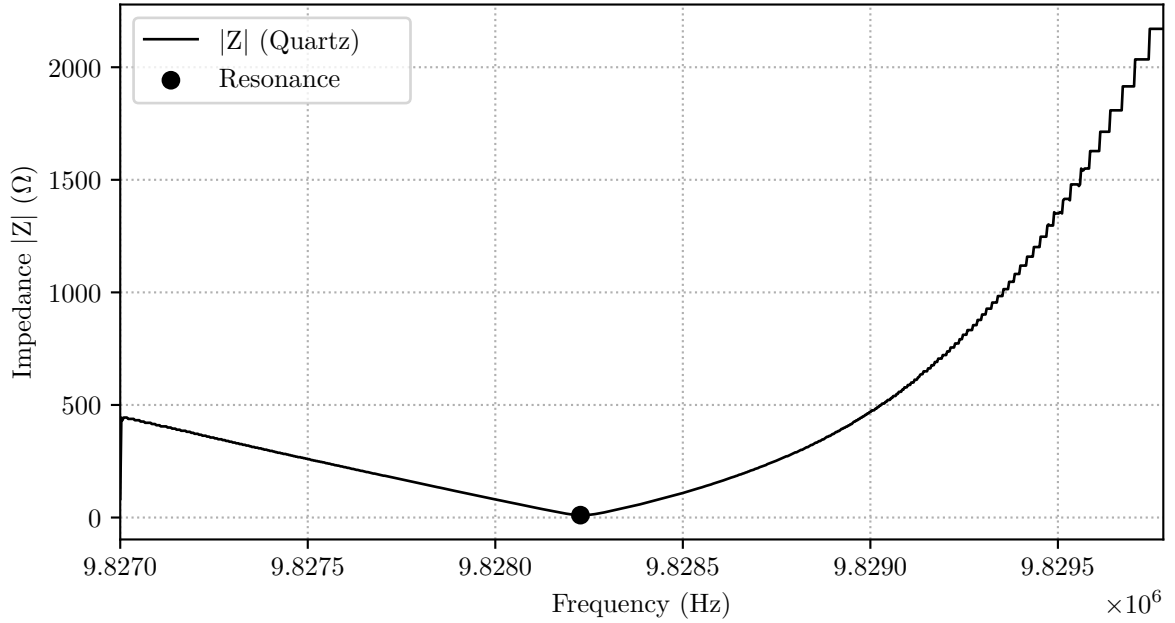


Figure 4: Measured impedance of a quartz resonator as a function of frequency.

Figure 4 shows the measured impedance magnitude $|Z|$ of the quartz resonator versus frequency. A distinct minimum of the impedance occurs at the resonance frequency $f_0 \approx 9.828227$ MHz.

At resonance, the inductive and capacitive reactances of the quartz cancel each other,

leaving only the small series resistance. As a result, the impedance reaches its lowest value. Away from resonance, reactive effects dominate, causing the impedance to increase rapidly [5].

3.5.5

The unknown component is an inductance. This is evident from the impedance behavior as seen in fig. 5. Decreasing at low frequencies and increasing linearly at high frequencies. At the intersection point where $|Z| = Z_0 = 50 \Omega$, the frequency is $f \approx 1.8 \times 10^7$ Hz.

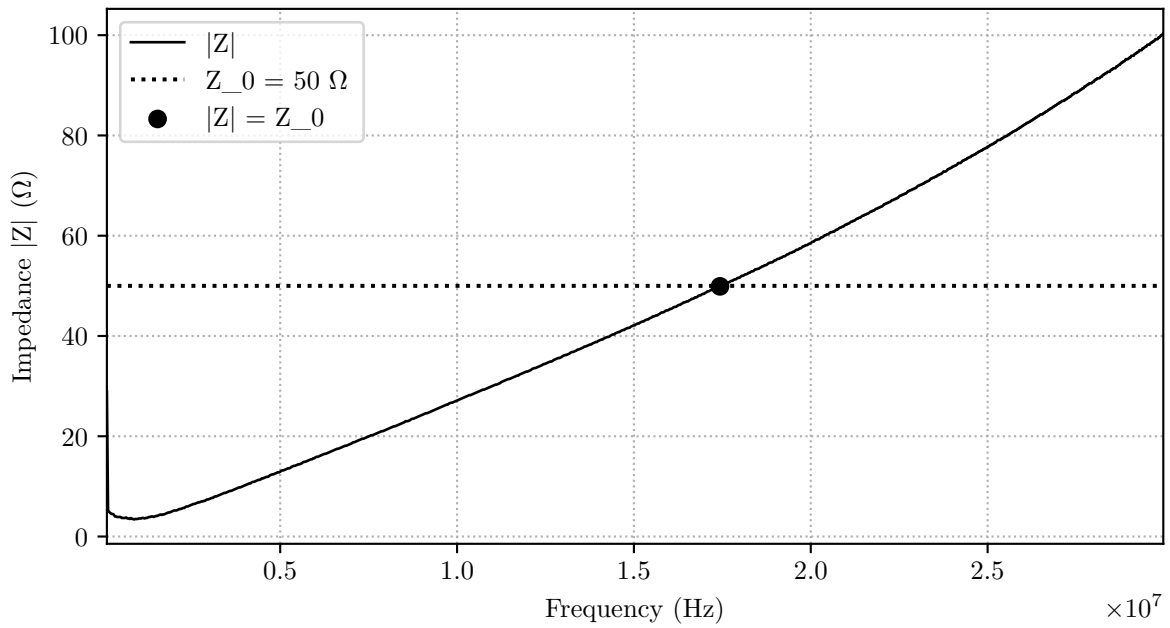


Figure 5: Measured impedance of an unknown part as a function of frequency.

The measured series resistance is 1Ω . To compare this value to the theoretical value the equation for our reflection at -0.18 dB and 90 degree needs to be evaluated [2]:

$$r = \frac{Z_L - Z_0}{Z_L + Z_0} \Leftrightarrow Z_L = Z_0 \frac{1 + r}{1 - r} \approx 1 + j 50 \Omega$$

$$L = \frac{X_L}{2\pi f} = \frac{50}{2\pi \cdot 17.434 \text{ MHz}} \approx 456,44 \text{ nH}$$

This result matches the measurement very well.

3.5.6

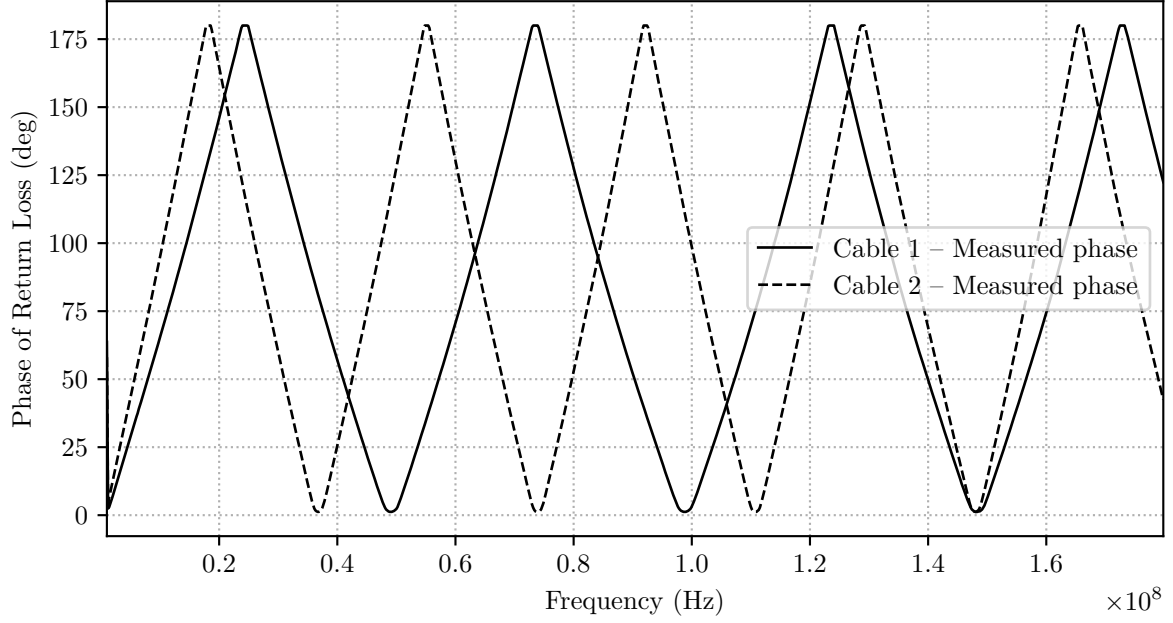


Figure 6: Measured phase of return loss of two different cables with different lengths a function of frequency.

From the formula of the reflection coefficient $r = r_L \cdot e^{-j2\beta l}$ and the fact that we can distinctly have maxima at a phase of 180 degree as seen in fig. 6, $-j2\beta l$ is set to π . in addition it is known that $\beta = \frac{2\pi}{\lambda}$ which results in a length of $l = \frac{\lambda}{4}$. After considering a velocity factor of 0.66 and the relation $\lambda_0 = \frac{c_0}{f}$ to connect the frequency to the wave length the resulting equation to calculate the lengths of the cables is $l = \frac{c_0 \cdot VF}{4 \cdot f}$ [1]. Cable 1 has the first peak at 24 339 327 Hz leading to a length of 2.7 m. Cable 2 has a peak at 18 359 836 Hz thus a length of 2.03 m.

3.5.7

Figure 7 shows the return loss of a matching network. A resonance can be observed at 14.238 MHz with a loss of -25.51 dB. The SWR is defined as $\frac{1+\rho}{1-\rho}$ [2]. With the measured return loss the SWR is 1.11.

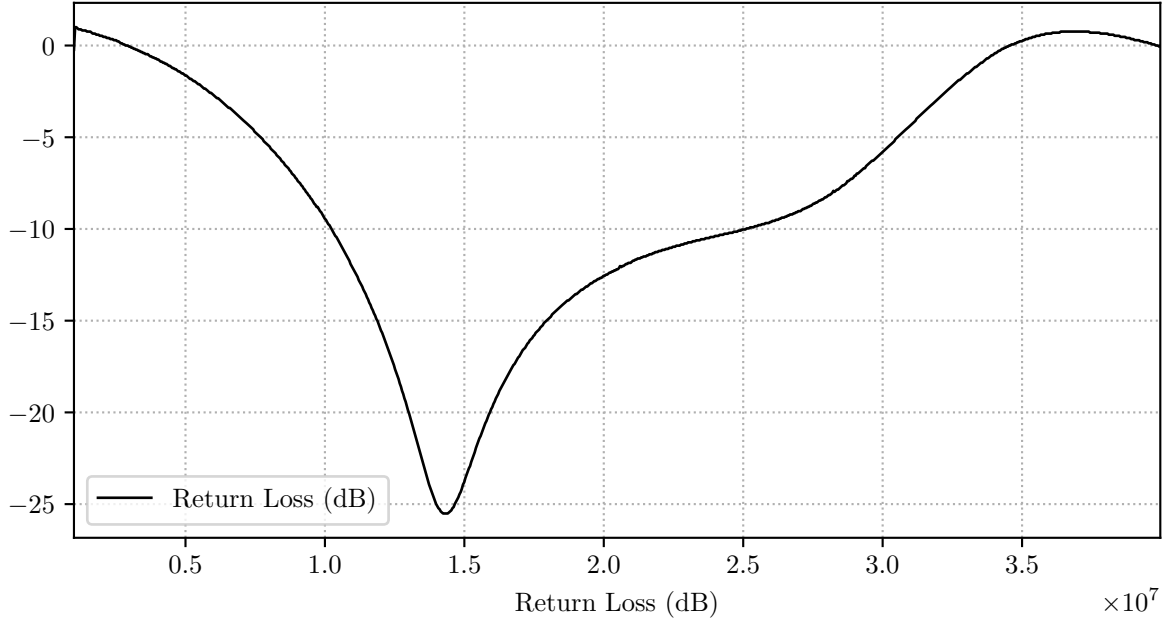


Figure 7: Measured Return Loss of Matching Network as a function of frequency.

3.5.8

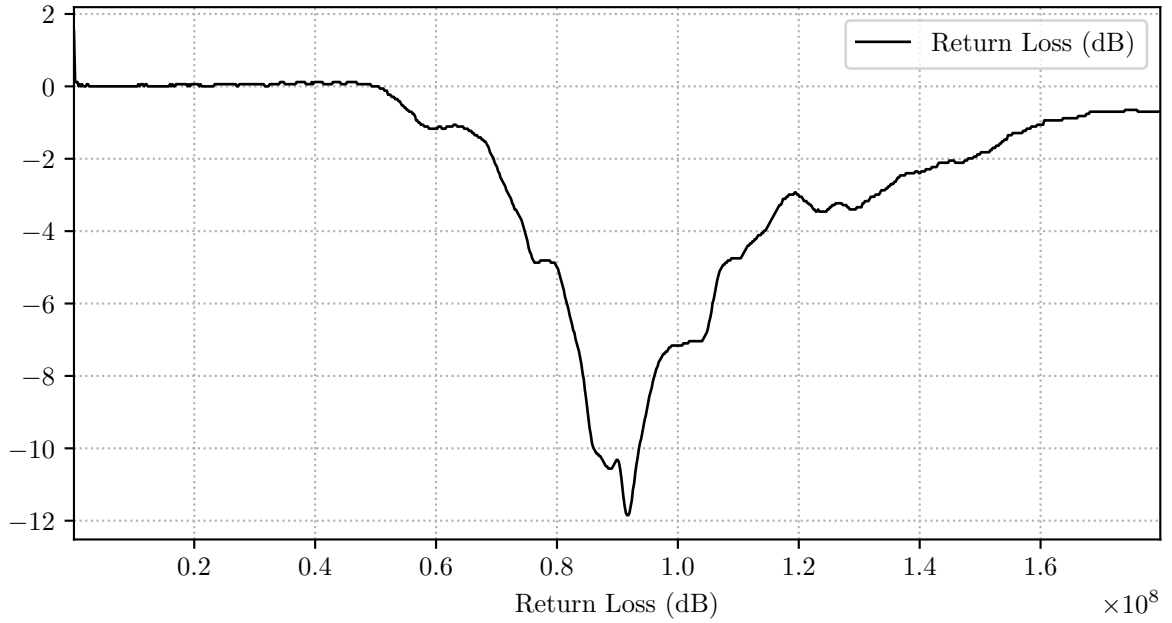


Figure 8: Measured return loss of a antenna as a function of frequency.

Figure 8 shows a minimum in the reflection coefficient at approximately 91.9 MHz, where the magnitude is measured as -11.85 dB. Substituting this value into the equation below gives the standing-wave ratio (SWR) [2]:

$$\text{SWR} = \frac{1 + 10^{\left(\frac{|S_{11}|_{\text{dB}}}{20}\right)}}{1 - 10^{\left(\frac{|S_{11}|_{\text{dB}}}{20}\right)}} = 1.69$$

The impedance can be determined just like shown in chapter 3.5.5, the reflection coefficient phase at resonance was measured at 152.5° . Thus the calculated value results in $Z_L = 30.78 + j7.75\Omega$. The resonance frequency lies within the FM broadcast band (87–108 MHz), the antenna is suitable for use in this frequency range.

3.5.9

For a monopole antenna, the approximate length is given by the quarter-wavelength rule $l = \frac{\lambda}{4}$. Substituting the free-space wavelength relation $\lambda = c_0/f$ yields:

$$l = \frac{c_0}{4f} = \frac{c_0}{4 \cdot 91.601 \text{ MHz}} = 0.818 \text{ m}$$

3.5.10

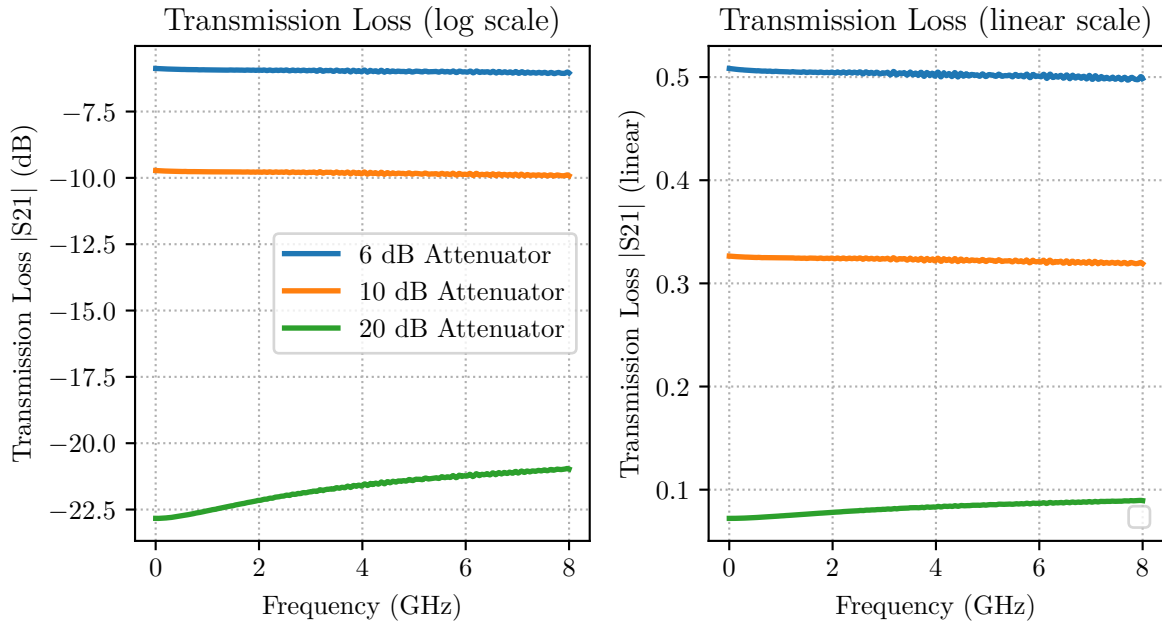


Figure 9: Transmission loss of three different attenuators

3.5.11

All three attenuators can be used for the entire frequency range of 8 GHz as can be seen in fig. 9. The cutoff frequency was given on the elements and is much higher at 18 GHz.

3.5.12

Since the ground creates a nearly perfect reflection due to having a impedance of zero, the signal passes through the attenuator two times resulting in a doubled damping thus a doubled return loss: 12 dB, 20 dB, 40 dB.

3.5.13

In figure 10 the center frequency of the LC band-pass can be determined at 8.502 MHz with a value of -0.18 dB. The return loss is measured at ~ 37 dB. The 3 dB threshold in transmission is measured at 7495021 Hz.

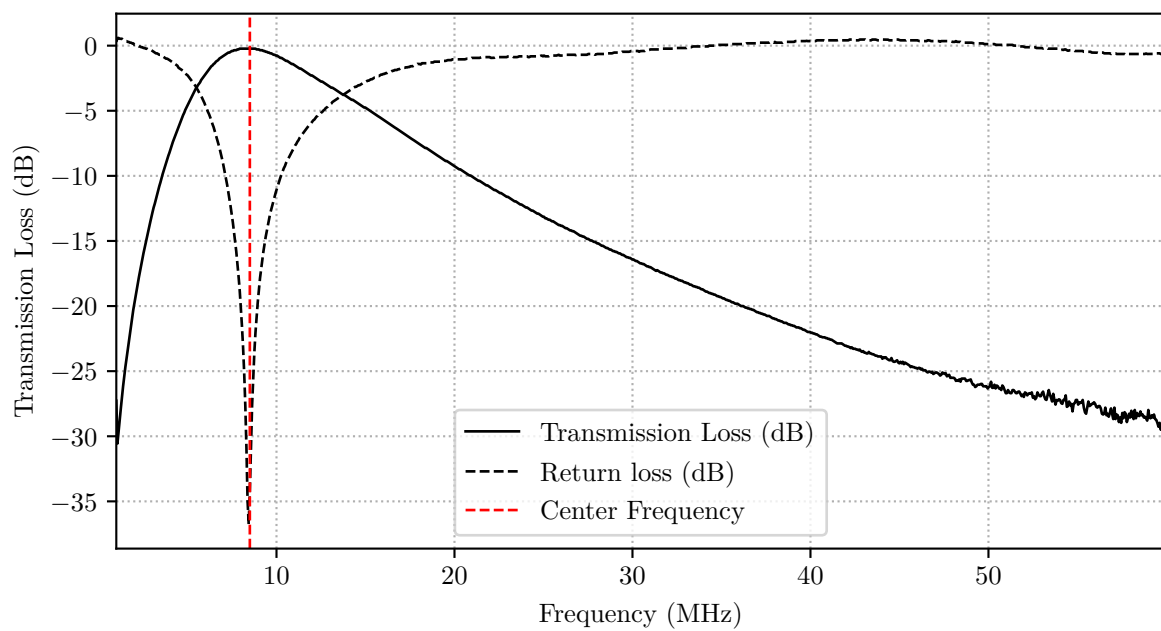


Figure 10: Measured LC Band-Pass Filter Transmission Loss a function of frequency.

3.5.14

The 6 dB bandwidth of the IF filter is ~ 270 kHz as can be seen in fig. 11. It is important to use double the 3 dB threshold because the IF filter only has one filter stage compared with the LC filter.

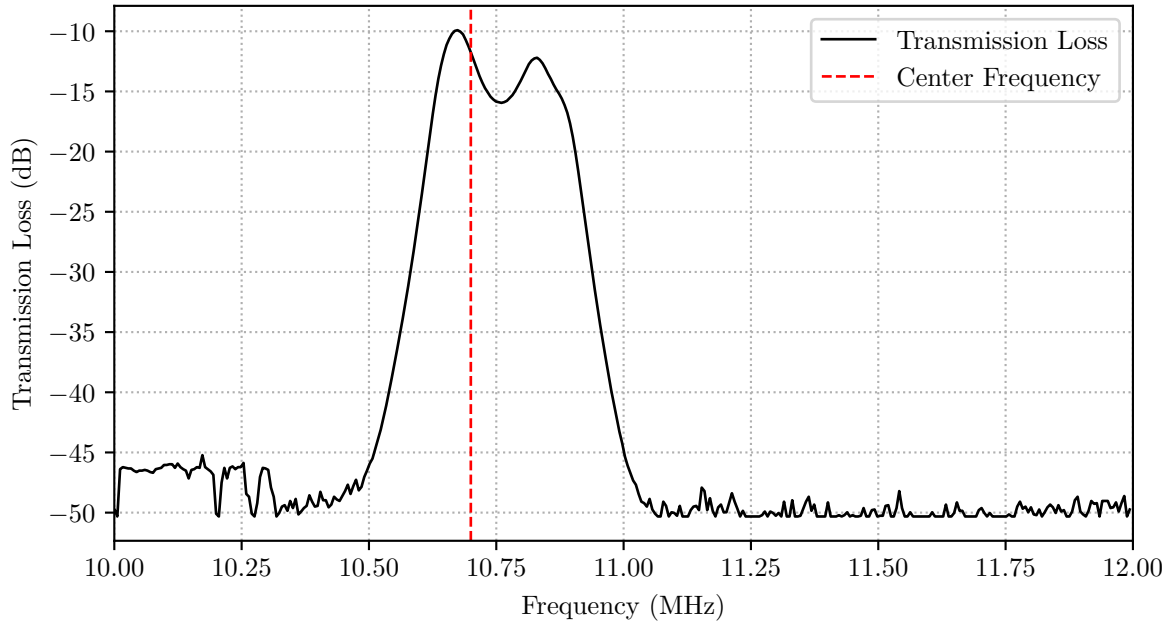


Figure 11: Measured IF Filter Transmission Loss as a function of frequency.

3.5.15

At the center frequency, the LC band-pass filter shows significantly lower insertion loss than the IF filter. The LC filter exhibits a transmission coefficient of -0.18 dB, whereas the IF filter shows -11 dB. At twice the center frequency, where strong attenuation is desired the LC filter attenuates the signal by only -6 dB, while the IF filter provides much stronger suppression. Consequently, the IF filter offers superior out-of-band rejection, whereas the LC band-pass filter provides better transmission at the center frequency.

References

- [1] Agilent Technologies. *Understanding the Fundamental Principles of Vector Network Analysis*. Agilent Technologies, 2007. Application Note.
- [2] Axel Bangert. *Praktikum Hochfrequenz-Schaltungstechnik*. Kassel.
- [3] E. Böhmer, D. Ehrhardt, and W. Oberschelp. *Elemente der angewandten Elektronik: Kompendium für Ausbildung und Beruf*. 17., korrigierte und verbesserte Auflage, 2018.
- [4] National Instruments. *HF-Komponenten unter der Mess-Lupe*. National Instruments, 2010. Fachartikel.
- [5] Ulrich Tietze, Christoph Schenk, and Eberhard Gamm. *Halbleiter-Schaltungstechnik*. Springer-Verlag, Berlin, Heidelberg, 13 edition, 2018.